

Variational approach to continuum orbitals in a spline basis: an application to H_2^+ photoionization

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A variational approach for the determination of continuum orbitals, recently proposed in the one-dimensional case, is generalized to the multicenter molecular problem. A test application is performed on the H_2^+ molecule in the one-center approximation, with a radial basis of B -spline functions. The resulting eigenvalue equation is solved by inverse subspace iteration giving a full set of K -matrix normalized solutions at each prefixed energy. Excellent numerical stability allows expansion up to $l=20$, giving very satisfactory agreement with exact results. The potential of a multicenter LCAO-type implementation is pointed out.

1. Introduction

Molecular photoionization is receiving increasing interest in the last years, as it provides detailed information both on the dynamics of the collision process and the intrinsic properties of the initial and final target states, which are directly related to the associated observables: total and partial cross sections and angular distributions. A large amount of experimental data on complex molecular systems, which is becoming increasingly available through the use of synchrotron radiation, calls for a corresponding level of theoretical description. In fact, the interpretation of the observed data, which are often beautifully structured, is not easily amenable to simple qualitative arguments, so that much of the available information is lost in the absence of theoretical interpretations.

The first purpose in photoionization properties calculations is a good representation of the continuum wave function. Basically, this can be performed by direct numerical integration of the differential equation or by expansion of the continuum functions as finite linear combinations of basis orbitals. Although much work has been done for small systems by a variety of approaches [1], only a few appear to be widely applicable to the multicenter problem posed by complex polyatomic targets, notably the Stieltjes imaging [2] and the continuum multiple scattering method (CMSM) [3], each suffering from significant limitations, so that a satisfactory solution of the continuum problem is still not available. The limitations of the Stieltjes imaging approach stem from the lack of asymptotic information. This prevents the extraction of the asymmetry parameter β as well as unambiguous separation of partial cross section contributions beyond the single-channel approximation. Moreover, the smoothing of the discrete pseudo-spectrum by the Stieltjes reconstruction makes it very difficult to treat narrow resonances, and basically limits the approach to the low-energy range. Most deficiencies have been recently overcome by the use of suitable oscillating functions [4], which allow the extraction of the full K -matrix, at the expense of a quite elaborate numerical treatment.

The CMSM is a natural extension of the multiple scattering bound-state formulation by Johnson [5], which allows the treatment of truly complex systems over a wide energy range. The use of the muffin-tin approximation to the potential, which is central to the approach, introduces, however, severe errors at the quantitative level, so that the scheme has been virtually abandoned in the bound-state problem, in favor of the LCAO approximation.

Furthermore, the original approach is a pure single particle one, so that resonances due to channel coupling

cannot be treated. As a matter of fact, attempts to lift the muffin-tin approximation or to introduce coupling between different channels has not met significant success.

Recently, a variational algorithm, based on a pure L^2 formulation, has been proposed for the one-dimensional atomic case, its numerical accuracy and stability was demonstrated [6], and a successful application to resonances in helium photoionization has been performed [7]. The approach combines the use of a spline function basis, pioneered by Shore several years ago [8–11], and a new variational principle, which entails the diagonalization of the matrix $A = H - ES$, obtained in the given basis. Because splines vanish identically outside a closed volume, the method amounts to the solution of the Schrödinger equation within a finite volume, large enough to reach the asymptotic region, so that it is then trivial to extract the full K -matrix normalized wave function by fitting the asymptotic part to a suitable combination of the regular and irregular asymptotic solutions.

The novelty of the approach is that it is variational, it does not need imposing any condition on the boundary, it is purely algebraic and extremely simple. It appears, therefore, ideally suited to an implementation in the multicenter molecular case.

Because of the restriction to a finite volume (a sphere) the approach bears some similarity with the R -matrix method [1]. The main difference is that the present scheme does not require imposing special boundary conditions, which severely restricts the class of basis functions available. Also the philosophy of the R -matrix approach is to extract full energy-independent information available from the basis by a single complete Hamiltonian matrix diagonalization, which makes it computationally trivial to construct the solution at each energy. As the full diagonalization problem is computationally very expensive, this strictly restricts the dimension of the basis which can be exploited in the R -matrix formalism.

On the contrary, the present approach requires only very few eigenvectors which have, however, to be extracted separately at each selected energy.

In the present work, we have considered an extension of the approach to the H_2^+ problem, for which accurate results for comparison have become recently available. A simple one-center expansion has been adopted, both for the bound and the continuum states, for computational convenience.

It is shown that the nontrivial problem posed by the coupling of different partial waves and the need of extracting a full set of independent solutions at each energy can be satisfactorily solved by applying the subspace inverse iteration approach.

The results show fast convergence and excellent agreement with those recently obtained by Richards and Larkins [12], indicating the potential of the present approach in the molecular case. In fact, no additional problems are anticipated for the use of an LCAO-type expansion, apart from the increased effort in computing the relevant matrix elements. In view of the recent advances in numerical integrations schemes [13–15] for the corresponding bound-state problem, this does not appear at present an intractable problem, at least in the case of a local potential or even in simple extensions like the time-dependent local density approach [16,17].

2. Theoretical method

The present approach is based on two separate aspects, the variational principle and the use of a spline basis set, which, although interrelated, are essentially independent.

2.1. Variational approach

In a recent paper, Froese Fischer and Idrees [6] proposed a generalization of the Rayleigh–Ritz–Galerkin approach for bound states to the determination of continuum wave functions. In the bound state problem, approximate solutions of the equation $H\Phi = E\Phi$ by expansion in basis set $\Phi = \sum_i c_i B_i$ is obtained by requiring stationarity of the expectation value,

$$\delta \frac{\langle \Phi | H | \Phi \rangle}{\langle \Phi | \Phi \rangle} = 0. \quad (1)$$

This is equivalent to the Galerkin equations,

$$b_i = \langle B_i | H - E | \Phi \rangle = 0, \quad i = 1, \dots, n, \quad (2)$$

which lead to the algebraic eigenvalue problem $(H - ES)c = 0$, determining eigenvalues and corresponding eigenvectors.

In the continuous spectrum the energy is fixed, so that the equation $Ac = 0$, $A(E) = H - ES$, in general does not admit nontrivial solutions $c \neq 0$. So it is proposed [6] to take as approximate solution the eigenvector c relative to the eigenvalue closest to zero (i.e., of minimum modulus). In this way, the expectation value $\langle \Phi | H - E | \Phi \rangle$ will be minimized, with the constraint $\sum_i |c_i|^2 = 1$. Intuitively, this condition is related to the minimization of the residual $\sum_i |b_i|^2$. In fact, variation of the functional (where the asterisk represents complex conjugate)

$$\sum_i b_i^* b_i - \lambda \sum_i c_i^* c_i, \quad (3)$$

leads to the eigenvalue equation

$$A^+ A c = \lambda c. \quad (4)$$

Although eq. (4) is different from

$$A c = a c, \quad (5)$$

we can easily observe that

$$A c = a c \quad \text{implies} \quad \langle c, A^+ A c \rangle = a^* a, \quad (6)$$

so that if the minimum modulus eigenvalue of A is well separated from the others, as we have always verified when the basis is flexible enough to provide an accurate solution, then the corresponding eigenvector should be close to the eigenvector of $A^+ A$ relative to its minimum eigenvalue, i.e., to the minimum value of the residual $\sum_i b_i^* b_i$. Both eqs. (4) and (5), as well as a number of other formulations corresponding to the choice of different norms, are currently investigated. It appears that although the equation $A^+ A c = \lambda c$ gives a superior accuracy for a fixed basis, the simpler equation $A c = a c$ is satisfactory, and the major factor influencing the overall accuracy obtainable is the flexibility of the basis. The simpler equation has been, therefore, adopted in the present investigation.

Due to the lack of a boundary condition $\Phi(R) = 0$, the matrix A is not Hermitian, so that eigenvalues and eigenvectors are generally complex, as has been verified in the present case. As the matrix is real, complex eigenvalues and eigenvectors necessarily occur in conjugate pairs (a, c) and (a^*, c^*) . This was never observed for the minimum modulus eigenvalue in the one-dimension case because only one solution is obtained, so that the relevant eigenvalue is nondegenerate (in modulus). In the case of n coupled partial waves, always n eigenvalues are found with moduli much smaller than the others (close to zero), corresponding to the n independent solutions sought. In this case, conjugate pairs are frequently encountered. One can avoid complex solutions taking linear combinations $c_1 = c + c^*$, $c_2 = i(c - c^*)$, which corresponds to taking linear combinations of the degenerate solutions of the continuum problem. Moreover, as c_1 and c_2 span the same invariant subspace, one can entirely avoid complex algebra by solving for two arbitrary linearly independent vectors in the same space.

Because of the large separation between the moduli of the n lowest eigenvalues and the rest, inverse iteration [18,19] applied to a subspace of dimension n provides a very effective means of extracting the required vectors.

2.2. Spline basis

The limitation of the configuration space to a finite volume, which gives the L^2 characteristic to the approach, strongly suggests the use of basis functions with compact support, (i.e., vanishing outside a closed interval), although, this is not strictly necessary. B -splines [20] are ideally suited for this purpose. In the present work, products of radial B -splines times spherical harmonics have been employed,

$$\Phi_{ilm} = \frac{1}{r} B_i(r) Y_{lm}(\theta, \varphi). \quad (7)$$

The required matrix elements $S_{ij} = \langle B_i, B_j \rangle$ and $H_{ij} = \langle B_i, HB_j \rangle$, relative to the one-electron electrostatic Hamiltonian,

$$H = -\frac{1}{2}\Delta - \frac{1}{|R_1 - r|} - \frac{1}{|R_2 - r|}, \quad (8)$$

are easily reduced to products of analytical angular integrals times radial integrals.

Splines are piecewise polynomials $S(x)$ chosen to satisfy fixed constraints. Given an interval $[a, b]$, divided into m subintervals by $m-1$ internal knots k_i

$$a < k_1 < k_2 < \dots < k_{m-1} < b, \quad (9)$$

$S(x)$ is a polynomial of degree n on each subinterval, and it is of class C^{n-1} over all $[a, b]$. This is realized by requesting continuity of the function and its $n-1$ derivatives in each knot,

$$S^+(k_i) = S^-(k_i),$$

...

$$S^{(n-1)+}(k_i) = S^{(n-1)-}(k_i), \quad i = 1, \dots, m-1. \quad (10)$$

It follows that one has $m(n+1)$ free parameters and $n(m-1)$ constraints, so that the dimension of the spline basis is $m+n$. This means that increasing the order or the number of the intervals by one just adds one basis function. A basis for the vector space spanned by the $n+m$ independent splines can be found in several ways. The B -spline basis is defined by the requirement of minimal support (i.e., to be nonzero over the smallest interval), which leads to basis functions being nonzero over at most $n+1$ adjacent intervals.

It is noteworthy that the choice of coincident knots gives the possibility to modify the continuity request at the knot, according to $c = n+1 - nk$, where c is the number of continuity conditions, and nk the number of coincident knots. Thus a $(n+1)$ -fold knot corresponds to no continuity conditions, while simple knots correspond to n continuity conditions. We choose maximum continuity at the interior knots, but no constraints at the endpoints, apart from the condition $f(0) = 0$, which is satisfied by simply deleting the first spline from the basis. The symmetrical condition $f(R) = 0$ at the end of the interval is employed in the bound-state case, deleting the last spline.

The flexibility of the polynomial basis allows a very good representation of the oscillatory nature of continuum orbitals as well as of the more slowly varying bound states. It allows the expansion of the basis set towards completeness by enlarging the degree or reducing the interval of support, without complications due to linear dependence problems. In fact, smooth convergence to machine accuracy has been observed in the one-dimensional case.

An additional advantage is given by the banded structure of the resulting matrices, because of the zero overlap between functions with disjoint supports.

3. Computational details

Calculations were performed at the internuclear distance of $R = 2.0$ au. Splines of order 10 have been em-

ployed. A fixed mesh allows the matrices S and H to be computed only once for all energies, greatly reducing the computational effort required at each energy, now simply limited to the diagonalization step. After some tests an interval of 20 au, and a uniform mesh with step $h=0.125$ were chosen. This allows good accuracy to be retained up to a $k=\sqrt{2E}$ value around 4, i.e., photoelectron energies up to 8 au can be explored. The maximum k value obtainable for a preset accuracy was found to be linearly varying with inverse step size in the Coulomb problem.

A linear mesh was selected as at sufficiently large energies (about $k > 1$) the stepsize required is largely determined by the quick oscillations of the continuum orbitals, which are rather weakly influenced even by the singular potential at the origin. Radial integrals are evaluated by Gaussian integration of order 12, and are essentially exact.

For the solution of the bound-state problem, as well as for tests on the eigenvalue spectrum of the A matrix, standard algorithms have been employed [21,22], while for the continuum calculations reported, subspace inverse iteration has been used [18,19].

Basically, one starts with n independent zero order vectors x_i^0 , at each iteration the system

$$A y_i^{k+1} = x_i^k, \quad (11)$$

is solved, and the y_i^{k+1} are symmetrically orthonormalized,

$$x_i^{k+1} = S^{-1/2} y_i^{k+1}. \quad (12)$$

The LU factorization of A is employed for the solution of the linear system. The iteration is stopped when the subspace distance between two successive iterations, defined as

$$d^2 = n - \sum_{ij} |\langle x_i^{k+1}, x_j^k \rangle|^2, \quad (13)$$

is less than a prefixed threshold (typically, $d^2 < 10^{-9} - 10^{-10}$ in the present case).

Initial vectors have been chosen as

$$c_{ilm,j}^0 = \frac{\delta_{lm,j}}{\|c\|}, \quad (14)$$

where j goes over the partial waves lm employed in the calculation. Because of $D_{\infty h}$ symmetry, only one partial wave for each l value is allowed in each symmetry, with only l values of the same parity allowed. We have employed an expansion up to a maximum of $l=20$, giving 10 independent channels.

Because of subspace iteration is done in real arithmetic, the n independent vectors so obtained are real. As already mentioned, they are not necessarily eigenvectors of A , but only a basis in the n -dimensional subspace spanned by the n minimum modulus eigenvectors of A . Any basis contains, however, the same informations, linear combinations being just related to different choices for the asymptotic boundary conditions. So K -matrix normalization is achieved by fitting the obtained solutions to linear combinations of the asymptotic regular and irregular solutions (Coulomb functions f_l and g_l [23]) at the last two interior knots of the interval, giving

$$\Phi'_n = \sum_{lm} \frac{1}{r} P_{lm,n}(r) Y_{lm}, \quad (15)$$

$$P_{lm,n} = CS_{lm,n} f_l(r) + CC_{lm,n} g_l(r), \quad (16)$$

$$K = CC^* CS^{-1}, \quad \Phi_n = \sum_n CS_{n',n}^{-1} \Phi'_{n'}, \quad (17)$$

$$\Phi_n(r) \rightarrow \sum_{lm} \frac{1}{r} (\delta_{lm,n} f_l + K_{lm,n} g_l) Y_{lm}. \quad (18)$$

The K -matrix so obtained is very nearly symmetric, as it is required for an exact solution [24]; it was nonetheless further symmetrized as $K = \frac{1}{2} (K + K^T)$.

The K -matrix gives informations about the coupling between different angular momenta due to the nonspherical molecular potential; it is a compact way to represent electron–molecule interaction. The corresponding eigenvalues

$$k_i = \tan \eta_i, \quad i = 1, \dots, n, \quad (19)$$

define the partial wave phase shifts η_i .

Photoionization properties have been calculated by Dill and Dehmer's formulas [3,25].

The differential cross section for the ejection of an electron into the continuum is given by

$$\frac{d\sigma}{d\Omega} = \frac{\sigma}{4\pi} [1 + \beta P_2(\cos \theta)], \quad (20)$$

where σ is the partial cross section and β the asymmetry parameter.

Resolving into separate contributions for each of the alternative values of the angular momentum (jt) transferred in the photoionization process, we obtain

$$\sigma = \sum_{jt} \sigma(jt), \quad (21)$$

$$\beta = \frac{1}{\sigma} \sum_{jt} \sigma(jt) \beta(jt), \quad (22)$$

$$\sigma(jt) = \frac{4}{3} \pi^2 \alpha h \nu \sum_l I(jt; ll), \quad (23)$$

$$\begin{aligned} \beta(jt) = & \left((2jt+1) \sum_l I(jt; ll) \right)^{-1} \{ (jt+2) I(jt; jt+1, jt+1) + (jt-1) I(jt; jt-1, jt-1) \\ & + 6 [jt(jt+1)]^{1/2} \operatorname{Re} [\exp(i(\sigma_{jt+1} - \sigma_{jt-1})) I(jt; jt+1, jt-1)] \}. \end{aligned} \quad (24)$$

The factor I represents the dynamical part of the photoionization process,

$$I(jt; ll') = \sum_{m, m_\gamma, m', m'_\gamma} D_{l'm'_\gamma m'_\gamma}^{-j\prime} D_{lm m_\gamma}^{-jt} \delta_{m-m_\gamma, m'-m'_\gamma}, \quad (25)$$

$$D_{lm_\gamma}^{-jt} = (-1)^{m_\gamma} (jt \ m - m_\gamma / 1 - m_\gamma, \ l m) D_{lm_\gamma}^{-j}, \quad (26)$$

$$D_{l'm_\gamma}^{-j\prime} = \sum_{l''} (I - iK)_{l''}^{-1} D_{l''m_\gamma}^{-j\prime}, \quad (27)$$

$$D_{l'm_\gamma}^{-j\prime} = \left(\frac{4}{3}\pi\right)^{1/2} \langle \Phi_{l'} | r Y_{lm_\gamma}(\hat{r}) | \Phi^0 \rangle, \quad (28)$$

with K -normalized continuum wave function $\Phi_{l'}$.

4. Results and discussion

The quality of the present one-center expansion may be judged by the bound-state energies reported in table 1. The convergence towards exact values rapidly increases with angular momentum, the largest differences being apparent for the lowest σ_g and σ_u states. This is tied to the cusps present in the LCAO expansion $1\sigma_{g,u} = 1s_1 \pm 1s_2$ which require large l values in a one-center expansion. Then the error drops by an order of magnitude going to the next excited state, and again to the next in σ_g , while it increases in σ_u . The latter is, however, due to the truncation of the wave function at $R=20$, and is apparent in all levels deriving from the $n=3$ levels of the united atom. For π_g or δ_g symmetries essentially exact results are obtained for the lowest level. With $l_{\max}=8$, the $1\sigma_g$ energy is in good agreement with the previous one-center value [26]. The final value, $l_{\max}=20$, gives an accuracy of 10^{-4} au on the ground-state energy, and is significantly more accurate than the well-known analytic represen-

Table 1

Bound states for the H_2^+ molecule (one-center approximation), $n=10$, $h=0.25$, $[0, R]=20$

Symmetry	$L_{\max}$	$-E$	$-E_{\text{exact}} [32]$	ΔE
σ_g	6	1.09994	1.10263	0.00269
	8 ^{a)}	1.10126		0.00137
	20 ^{a)}	1.10250		0.00013
		0.36050	0.36086	0.00036
		0.23573	0.23577	0.00004
σ_u	6	0.66559	0.66753	0.00194
		0.25503	0.25541	0.00038
		0.13611	0.13731	0.00120
π_g	6	0.22669	0.22669	0.0
		0.12558	0.12671	0.00113
δ_g	6	0.21273	0.21273	0.0
		0.11937	0.12101	0.00164

^{a)} $h=0.125$.

tation of James [27] or Guillemin–Zener [28]. As has been shown in ref. [29], the dipole matrix elements are strongly dependent on the details of the ground-state function, so the accuracy of the present representation is confirmed by the quality of the computed cross sections.

The continuum wave function accuracy can be tested from the phase shifts, reported in table 2. They provide a good representation of the rescattering of the initial electron angular momentum into a range of momenta due to the anisotropy of the molecular potential.

While the wave functions obtained by Richards and Larkins [12] in elliptical coordinates are essentially exact, the phaseshifts extracted from them are approximate, because of the neglect of the coupling of different angular momenta. That is, their phase shifts just reflect the diagonal elements of the CC matrix [eq. (16)] in their representation. For this reason, we have reported for comparison both full phaseshifts from K -matrix eigenvalues and the uncoupled diagonal K -matrix elements. The third row at each energy reports their values. At low energy, the effect of coupling is small, and the agreement of the full results with ref. [12] is very satisfactory. As the energy increases, the difference between coupled and uncoupled values gets larger, especially in the σ_u symmetry, and the values of Richards and Larkins become much closer to the latter. Because of the coupling, it appears probable that the present results are more accurate than the previous ones, despite the essentially exact wave function of the former study.

As eigenvalues, also K -matrix eigenvectors yield some information. In fact, according to Loomba et al. [30], if an eigenchannel is dominated by a particular orbital momentum, then the eigenchannel wave function has the characteristic angular pattern of the corresponding spherical harmonic.

As an example of the importance of angular momenta coupling we report in table 3 selected K -matrix eigenvectors from a smaller (five channels) calculation of the H_2^+ continuum wave functions. An expansion up to $l_{\max}=9$ has been used, which corresponds to a mixing of angular momenta 1, 3, 5, 7, 9 for the $D_{\infty h}$ symmetry. At low energy (0.6 au), we can see that the coupling is present but not very important; in fact eigenvectors 1, 2, 3 are dominated respectively by the c_1 , c_2 , c_3 components, so we can say that the characterizing orbital momenta for them are basically $l=1$ for the first, $l=3$ for the second and $l=5$ for the third. At higher energy, however, the coupling becomes stronger and the various components of an eigenchannel are no longer well separated, having comparable weights: it follows that it is difficult to recognize which orbital momentum characterizes which eigenvector.

Cross sections and asymmetry parameters are reported in table 4. Photoionization cross sections are essentially consistent with previous results [12]. Notably the agreement with the exact results of ref. [12] is excellent, the error being on the fourth significant figure. With $l_{\max}=9$, the values are systematically too low by about 1%

Table 2

Phase shifts for the H_2^+ molecule in units of π radians (one-center approximation), $n=10$, $h=0.125$, $[0, R]=20$, $l_{\max}=20$

Photoel. energy	$p\sigma_u$	$f\sigma_u$	$h\sigma_u$	$p\pi_u$	$f\pi_u$	$h\pi_u$
0.2	0.13976 ^{a)}	0.02904	0.00760	-0.14950	0.02032	0.00681
	0.13970 ^{b)}	0.02892	0.00734	-0.14946	0.02000	0.00658
	0.14030 ^{c)}	0.02922	0.00764	-0.14968	0.02023	0.00685
0.4	0.12205	0.03199	0.00889	-0.14923	0.02228	0.00787
	0.12159	0.03206	0.00865	-0.14912	0.02150	0.00774
	0.12203	0.03228	0.00885	-0.14926	0.02167	0.00792
0.6	0.10738	0.03429	0.00991	-0.14944	0.02399	0.00870
	0.10601	0.03507	0.00972	-0.14927	0.02275	0.00867
	0.10641	0.03518	0.00987	-0.14933	0.02288	0.00881
0.8	0.09516	0.03583	0.01078	-0.14985	0.02546	0.00940
	0.09220	0.03798	0.01066	-0.14964	0.02380	0.00949
	0.09264	0.03794	0.01076	-0.14962	0.02389	0.00959
1.0	0.08522	0.03637	0.01155	-0.15033	0.02674	0.01001
	0.07973	0.04084	0.01153	-0.15011	0.02468	0.01023
	0.08028	0.04060	0.01158	-0.15001	0.02474	0.01029
1.2	0.07767	0.03559	0.01221	-0.15082	0.02786	0.01056
	0.06831	0.04364	0.01233	-0.15060	0.02540	0.01092
	0.06904	0.04319	0.01233	-0.15043	0.02544	0.01094
1.4	0.07263	0.03320	0.01277	-0.15130	0.02883	0.01105
	0.05776	0.04639	0.01310	-0.15108	0.02599	0.01156
	0.05872	0.04571	0.01304	-0.15086	0.02602	0.01154
1.6	0.06990	0.02935	0.01317	-0.15174	0.02967	0.01150
	0.04912	0.04795	0.01384	-0.15154	0.02646	0.01218
	0.04918	0.04818	0.01372	-0.15128	0.02649	0.01211
1.8	0.02471	0.06892	0.01325	-0.15214	0.03040	0.01191
	0.03881	0.05179	0.01454	-0.15196	0.02681	0.01276
	0.04032	0.05061	0.01436	-0.15166	0.02686	0.01265
2.0	0.02046	0.06906	0.01241	-0.15250	0.03104	0.01228
	0.03024	0.05441	0.01523	-0.15234	0.02706	0.01332
	0.03205	0.05301	0.01498	-0.15202	0.02714	0.01316
3.0	-0.01213	0.07553	0.01858	-0.15363	0.03316	0.01369
	-0.00542	0.06653	0.01852	-0.15358	0.02701	0.01586
	-0.00231	0.06443	0.01780	-0.15332	0.02747	0.01545
4.0	-0.03429	0.08333	0.02068	-0.15386	0.03415	0.01452
	-0.03158	0.07637	0.02173	-0.15385	0.02529	0.01809
	-0.02813	0.07473	0.02035	-0.15388	0.02642	0.01742
6.0	-0.06324	0.09462	0.02457	-0.15272	0.03485	0.01508
	-0.06315	0.08627	0.02829	-0.15261	0.02185	0.01893
	-0.06335	0.09027	0.02511	-0.15354	0.02181	0.02075

^{a)} Eigenvalues of K matrix. ^{b)} Diagonal elements of K matrix. ^{c)} Ref. [12].

up to about 2 au, and within 2% up to 6 au. As already mentioned, a significant part of the disagreement may be due to an inadequate representation of the initial bound state.

The monotonic decrease of the total cross section with photoelectron energy is illustrated in fig. 1, while the relative contributions of each partial wave component, $\sigma_{lm}/\sigma_{\text{tot}}$ is reported in fig. 2.

Near the threshold, the dominant component is the $p\pi$ one, which decreases monotonically with increasing energy. All other components are very small at threshold, but their relative contributions increase quickly to a maximum, followed by a decrease at higher energies.

The $l=5$ components become significant only for photoelectron energies about 100 eV, while the higher ones

Table 3

K-matrix eigenvectors from a five channels $H_2^+ \pi_u$ continuum wave function calculation, $n=10$, $h=0.125$, $[0, R]=20$, $l_{\max}=9$

E	Eigenvector	c_1	c_2	c_3	c_4	c_5	$L^a)$
0.6	1	0.9994	-0.0339	0.0007	0.0000	0.0000	1
	2	-0.0326	-0.9665	-0.2529	-0.0285	-0.0019	3
	3	0.0086	0.2377	-0.8588	-0.4427	-0.0992	5
	4	-0.0031	-0.0842	0.4061	-0.7041	-0.5764	7
	5	-0.0012	-0.0331	0.1830	-0.5545	0.8111	9
1.2	1	0.9992	-0.0388	0.0012	0.0000	0.0000	1
	2	-0.0360	-0.9380	-0.3411	-0.0488	-0.0039	3
	3	0.0131	0.3124	-0.7847	-0.5184	-0.1333	5
	4	-0.0057	-0.1331	0.4623	-0.6209	-0.6189	7, 9
	5	-0.0025	-0.0573	0.2327	-0.5860	0.7740	7, 9
2.0	1	0.9995	-0.0325	0.0018	0.0000	0.0000	1
	2	-0.0285	-0.9014	-0.4256	-0.0744	-0.0070	3
	3	0.0136	0.3801	-0.7020	-0.5781	-0.1683	5, 7
	4	-0.0070	-0.1864	0.5007	-0.5419	-0.6488	5, 7, 9
	5	-0.0033	-0.0853	0.2745	-0.6055	0.7421	7, 9

^{a)} Dominant orbital momentum of the eigenchannel.

Table 4

Photoionization cross sections (Mb) and asymmetry parameter for H_2^+ , at same conditions as table 2

Photoelectron energy	σ_u		π_u		σ_{tot}	β
	(a)	(b)	(a)	(b)		
0.0	0.005211	0.005211	0.925606	0.926035	0.93082	1.23671
0.1	0.007387		0.690953		0.69834	1.34948
0.2	0.009777	0.009796	0.526954	0.527034	0.53673	1.40024
0.3	0.011863		0.409486		0.42135	1.44017
0.4	0.013494	0.013519	0.323611	0.323700	0.33710	1.47478
0.5	0.014678		0.259437		0.27412	1.50630
0.6	0.015464	0.015490	0.210766	0.210856	0.22623	1.53576
0.7	0.015927		0.173212		0.18914	1.56369
0.8	0.016125	0.016150	0.143900	0.143967	0.16002	1.59035
0.9	0.016123		0.120688		0.13681	1.61586
1.0	0.015962	0.015986	0.102140	0.102203	0.11810	1.64034
1.1	0.015686		0.087144		0.10283	1.66371
1.2	0.015323	0.015344	0.074910	0.074966	0.09023	1.68607
1.3	0.014896		0.064852		0.07975	1.70736
1.4	0.014429	0.014448	0.056504	0.056549	0.07093	1.72755
1.5	0.013932		0.049530		0.06346	1.74673
1.6	0.013417	0.013436	0.043664	0.043701	0.05708	1.76482
1.7	0.012897		0.038692		0.05159	1.78186
1.8	0.012375	0.012391	0.034456	0.034489	0.04683	1.79790
1.9	0.011856		0.030827		0.04268	1.81293
2.0	0.011346	0.011362	0.027698	0.027725	0.03904	1.82699
2.5	0.008993		0.017150		0.02614	1.88427
3.0	0.007039	0.007050	0.011440	0.011454	0.01848	1.92366
3.5	0.005472		0.008061		0.01353	1.95008
4.0	0.004236	0.004243	0.005917	0.005927	0.01015	1.96744
4.5	0.003268		0.004481		0.00775	1.97854
5.0	0.002515	0.002519	0.003476	0.003482	0.00599	1.98531
5.5	0.001932		0.002747		0.00468	1.98910
6.0	0.001482	0.001485	0.002205	0.002209	0.00369	1.99084
6.5	0.001137		0.001792		0.00293	1.99116
7.0	0.000874		0.001472		0.00235	1.99055
7.5	0.000673		0.001220		0.00189	1.98933
8.0	0.000521		0.001019		0.00154	1.98779

^{a)} Present results. ^{b)} Ref. [12].

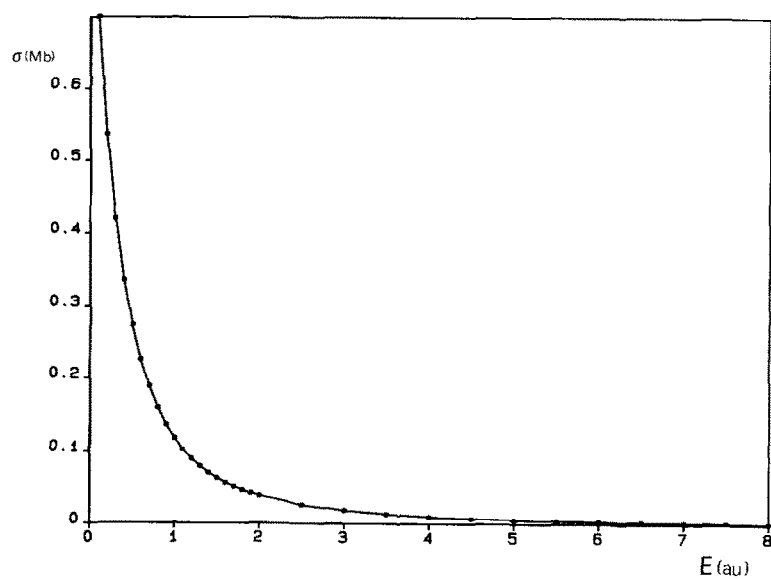


Fig. 1. Total cross section for photoionization of H_2^+ .

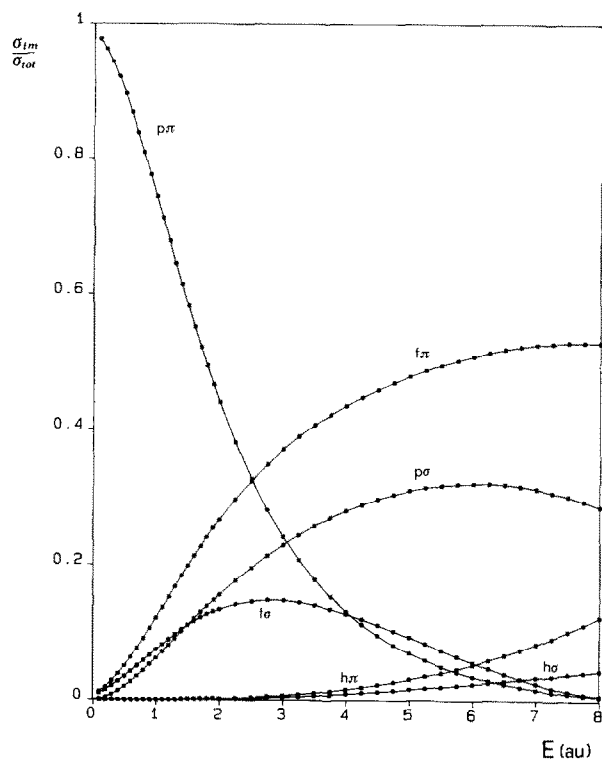


Fig. 2. Relative contribution of each component cross section σ_{lm} to the total cross section σ_{tot} .

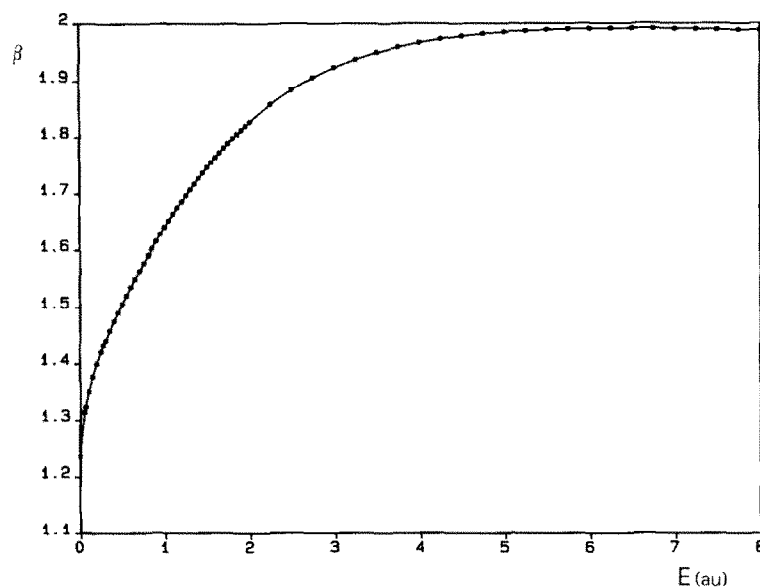


Fig. 3. Asymmetry parameter for photoionization of H_2^+ as a function of the photoelectron energy.

are negligible at the energies considered. The behavior observed is indeed very similar to that previously reported by Chapman and Hayes [31].

The pattern shown by the asymmetry parameter β as a function of photoelectron energy, which is reported in fig. 3, is in excellent agreement with that previously obtained in ref. [12], and is significantly different from the one-center result of Chapman and Hayes. It increases monotonically in the energy range 0–8 au from 1.24 to the asymptotic value of 2.

This shows that the disagreement between the two previous results cannot be attributed to the inadequacy of the one-center expansion, nor to the l -uncoupled approximation of ref. [12], but probably stems from an incorrect evaluation of the β parameter in ref. [31].

5. Concluding remarks

We have shown that the purely algebraic approach previously proposed for the one-dimensional problem can be successfully employed for the multicenter molecular case, and that a full set of linearly independent K -normalized solutions can be obtained at each energy. Let us summarize the principal points.

(a) The approach is purely L^2 , variational and very simple. Notably no boundary conditions, or necessity of supplementing the basis with asymptotic solutions are required. Alternative formulations which can give better convergence are being tested.

(b) A one-center approximation has been employed, but an LCAO approach is entirely feasible, at the expense of multicenter matrix elements. This does not appear to be too difficult, at least in a local potential scheme.

(c) The spline basis employed shows several advantages. B -splines are simple (polynomials), do not suffer from linear dependency problems, lead to rapid convergence with increasing order, which does not inordinately increase the dimension of the basis, and give rise to very sparse matrices.

(d) The core of the approach is the extraction of few eigenvectors of a given matrix relative to eigenvalues closest to zero, well separated from the rest. The matrix is unsymmetric, but can be made symmetric and positive definite at the expense of a matrix multiplication to give A^+A . An efficient diagonalization is achieved through

inverse subspace iteration. It appears possible that several large matrix techniques developed for the bound-state problem can be fruitfully generalized to the present contest. Indeed all knowledge available on the algebraic eigenvalue problem is immediately applicable. Also, perturbation theory may be easily employed, e.g., in the case of weak coupling between different channels.

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